Task 1 Report: Iris Exploratory Data Analysis

Healthcare + Data Science Project

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1 Problem Statement

The goal of this task is to perform exploratory data analysis (EDA) on the Iris dataset and identify meaningful trends in feature behavior, class-level separation, and potential outliers.

2 Objective

The analysis aims to:

- inspect dataset structure and summary statistics,
- understand feature distributions,
- analyze pairwise relationships across species,
- detect potential outliers.

3 Dataset Loading and Preprocessing

The Iris dataset is loaded using seaborn and validated for empty/null load states. Core preprocessing includes:

- schema inspection,
- shape and column checks,
- descriptive statistics for numerical columns.

4 Data Visualization and Exploration

4.1 Sepal Relationship by Species

Figure 1: Sepal length vs sepal width scatter plot with species coloring.

4.2 Pairwise Feature Space

Figure 2: Pairwise feature relationships across all Iris variables.

4.3 Feature Distributions

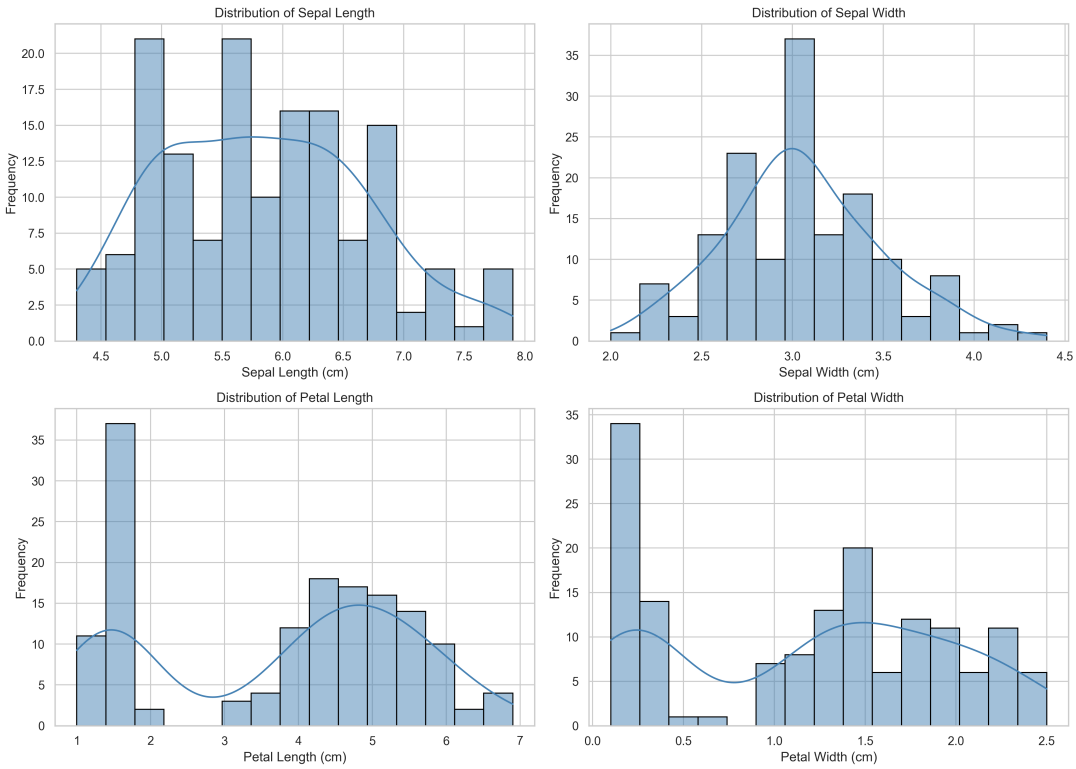

Figure 3: Histogram + KDE plots for all four numeric features.

4.4 Outlier Inspection by Species

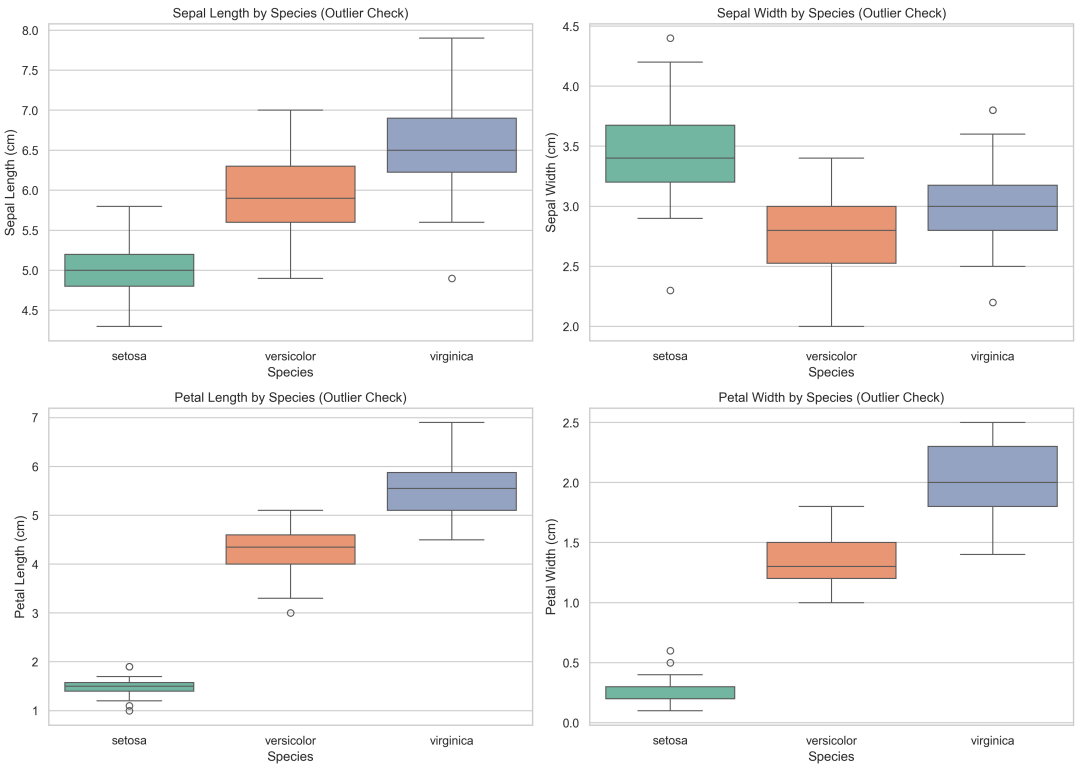

Figure 4: Species-wise boxplots for outlier and spread analysis.

5 Key Findings and Insights

- Petal-based features show stronger class separation than sepal-based features.
- Setosa appears clearly separable in most pairwise views.
- Versicolor and virginica overlap in some regions, indicating moderate class boundary ambiguity.
- Box plots reveal a few outlier points, but overall class trends remain consistent.

6 Conclusion

The EDA confirms that the Iris dataset contains strong visual signal for class discrimination, especially in petal features. These insights support downstream supervised learning tasks such as species classification.